OList Delivery ETA

Unreliable Delivery ETAs May Reduce Customer Trust and Therefore Future Purchases

Our Solution

Our solution aims to provide more reliable delivery estimates and improve customer confidence.

We developed a machine learning model for Olist employees to predict delivery time after order approval

In the following slides, we will explain how our model works and how it can support more transparent delivery expectations

What our solution provides:



More Reliable
Delivery Estimates

Fast Predictions
allow customers to feel reassured upon purchase

Tracking Order Times
to allow customers to be prepared for delivery

Overview of Our Process

Steps Taken:

Improved Feature Quality to maximize predicting power

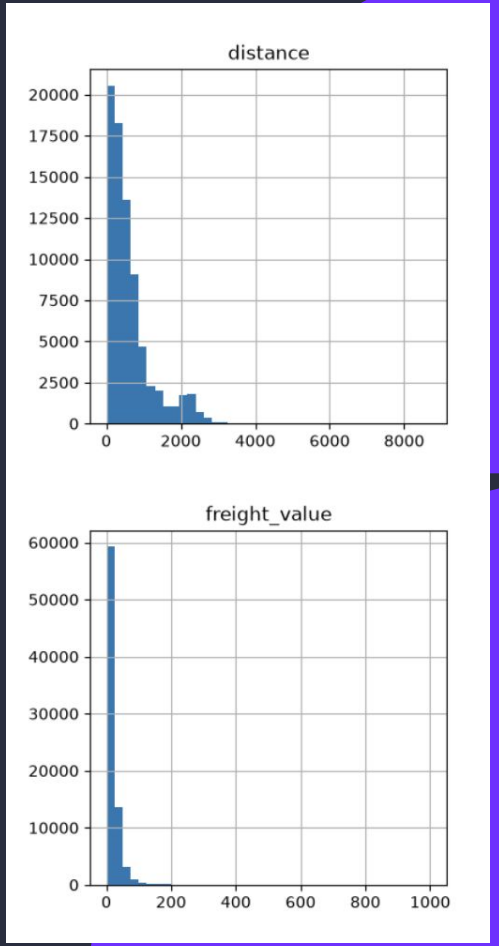
Determined which factors affect delivery times the most

Traded interpretability for higher prediction accuracy

Feature Overview and Selection

- Distance between buyer and seller determined as a major factor

- Freight cost as a proxy for product details

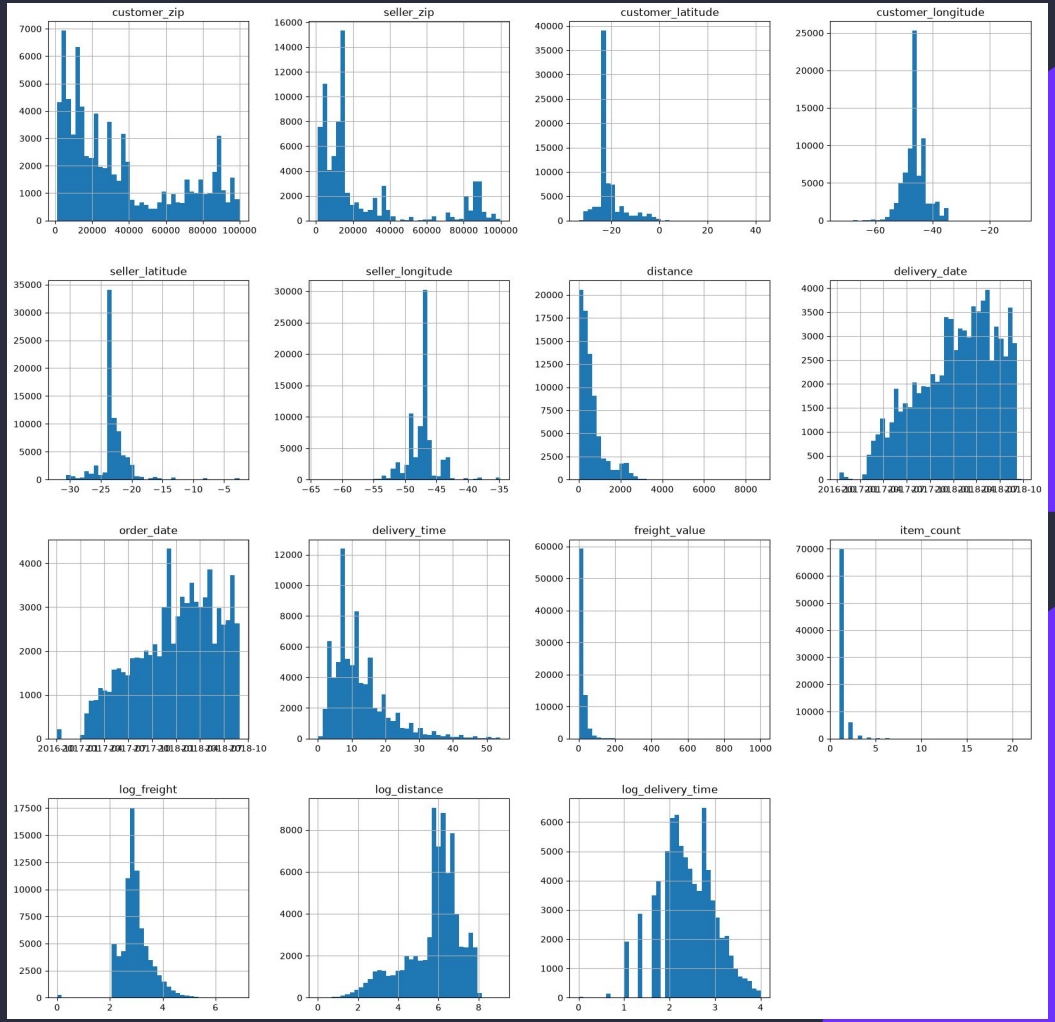


Decision Tree Baseline Model

Feature Distribution

- Distances and Freight values are heavily skewed

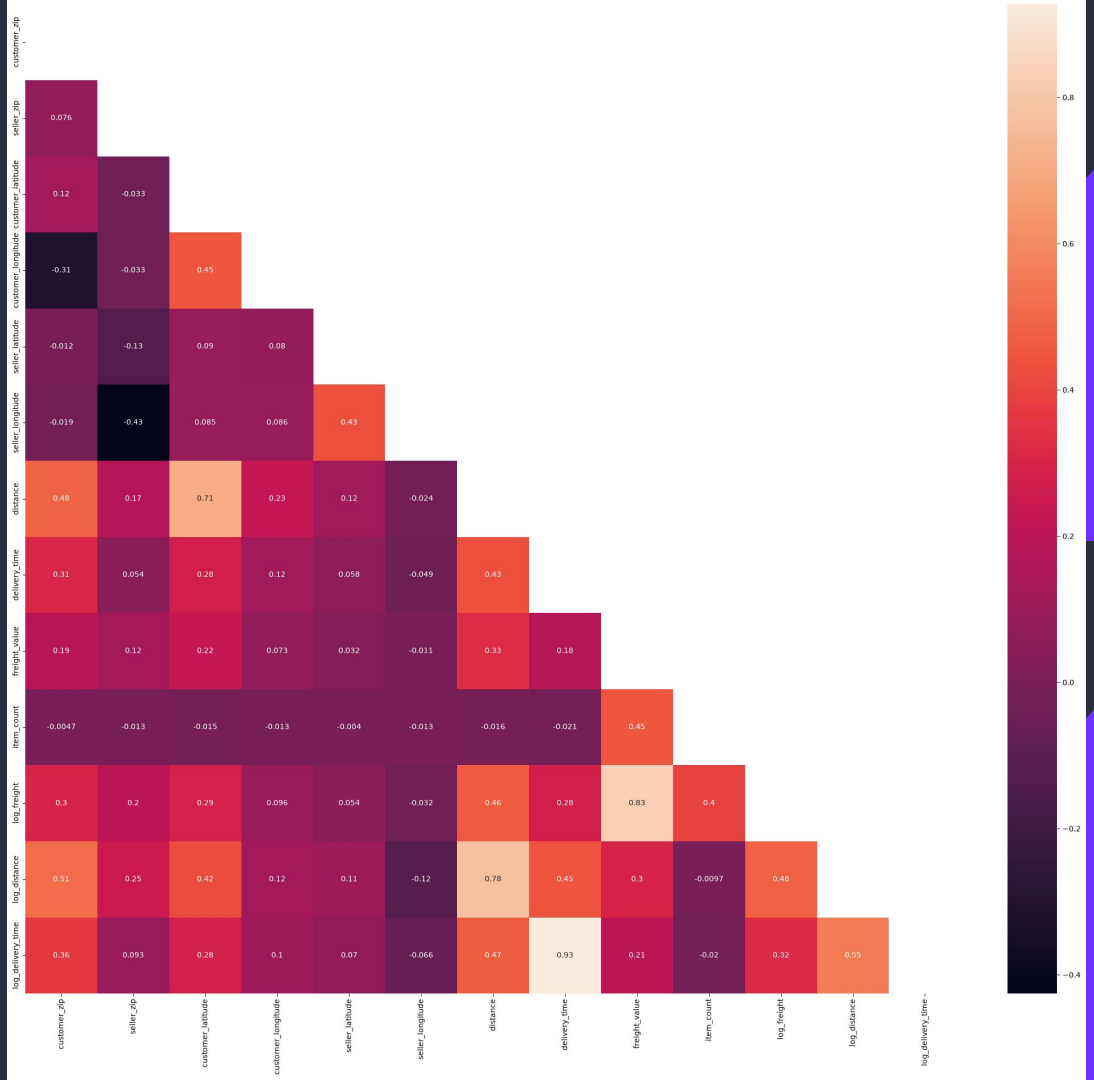
- Log taken to improve prediction accuracy



Feature Correlation

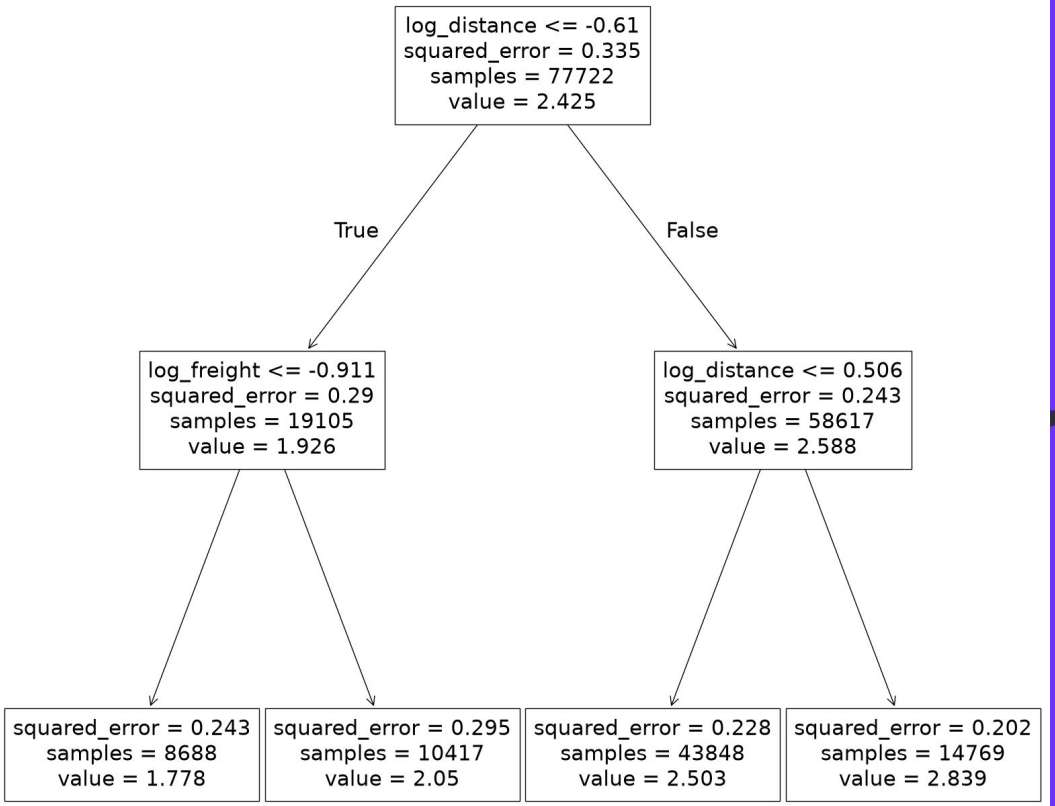
- Distance and Freight values used

- Zip, latitude and longitude not used to avoid reuse of features



Tree Model

- Model predictions were off by 5 days on average



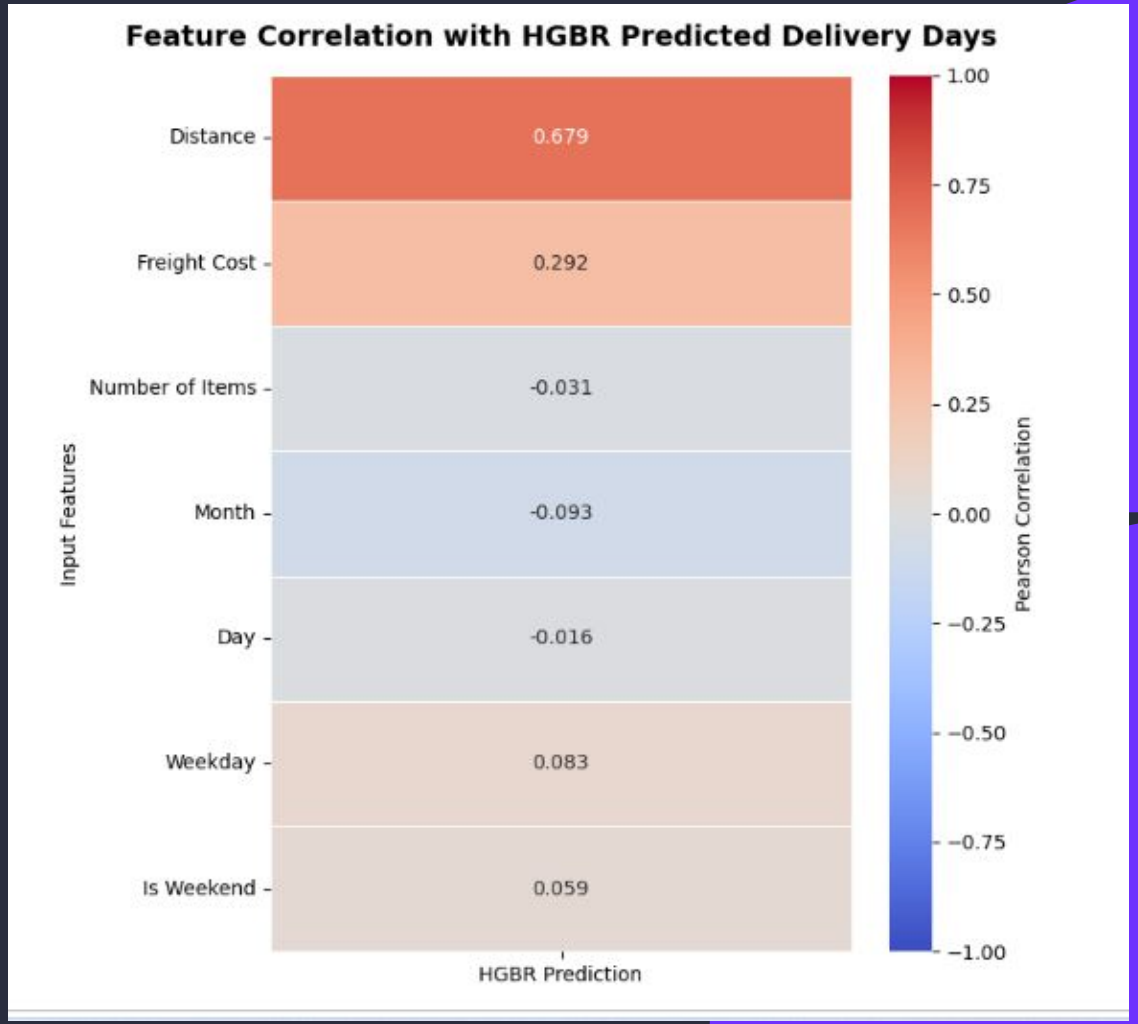
HGBR Model

HGBR Feature Correlation

As before:

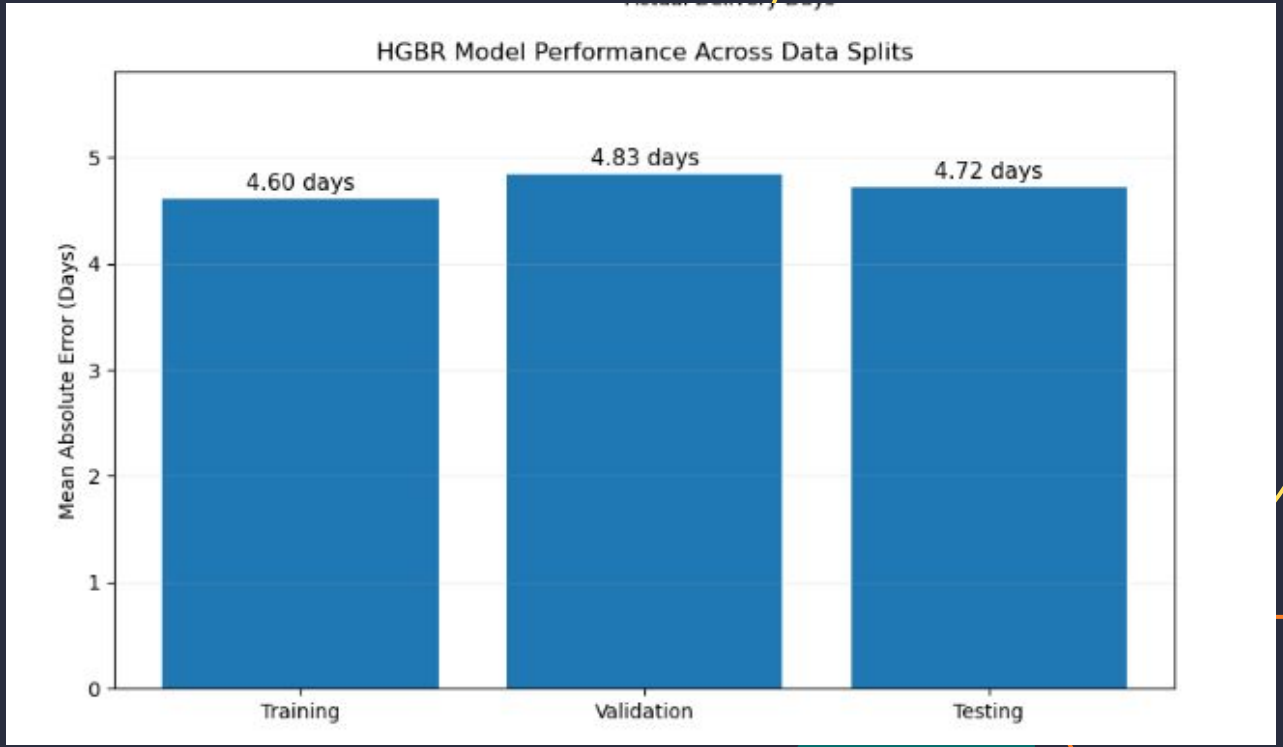
- Distance
- Freight Cost

- Item Count used to improve prediction accuracy



Model Performance

- To provide estimates which appear more reliable, **predictions** are **rounded up** to the **nearest whole day**.
- As a simple example, a prediction of **4.1 days** would **become 5 days**.



Selecting Our Model

Why We Chose HGBR Over Decision Tree

- **Lower Prediction Error:** HGBR achieved an MAE 0.5 days lower than the Decision Tree model.
- **Faster Predictions:** HGBR demonstrated faster prediction times in our testing.
- **Similar Rounded MAE:** Both models had an MAE of approximately 5 days when rounded up.
- **Better Accuracy:** HGBR provided more accurate delivery time predictions.
- **Final Decision:** We selected HGBR because it offered a **better balance** of prediction **accuracy** and **speed**.

Demo:



Next Steps

01.

Reduce the used features to streamline usage

02.

Should the model be forward-facing, select features which are available to customers at purchase time

03.

Streamline Model Choice to reduce inference latency to allow faster ETA predictions

04.

Work with Olist to further refine model requirements and potential improvements

Thank you